

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2077 Chaitra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Artificial Intelligence (Elective III) (CT78506)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is Turing Test and total turing test? Explain PEAS for a self-driving car. [4+4]
2. Assume you are given two jugs; a 4-gallon one and a 3-gallon one, and a pump which has unlimited water. How can you get exactly 2 gallons of water in the 4-gallon jug? Formalize the problems, write down Production Rules and draw Search Tree for water Jug problem. [2+2+4]
3. Devise an example to show how A* algorithm uses path cost and heuristic cost to generate best solution. [8]
4. Explain forward chaining giving suitable example. [8]
5. If it is sunny and warm day you will enjoy. If it is warm and pleasant day you will do strawberry picking. If it is raining no strawberry picking. If it is raining you will get wet. It is warm day. It is raining. It is sunny. Prove that "You will enjoy" using resolution by refutation. [8]
6. What are Frames and Semantic Net? Convert the given sentences in semantic Net: [2+6]
 - Tweety and Sweety are birds.
 - Tweety has a red beak.
 - Sweety is Tweety's child.
 - A crow is a bird.
 - Birds can fly.
 - Sparrow is a bird.
 - Sparrow has a wing.
7. You are locked on a deserted island. Mushrooms of various types grow widely all over the island but no other food is anywhere to be found. Some of the mushrooms have been determined as poisonous and others as not (Determined by your former companions' trial and error). You are the only one remaining on the island. You have the following data to consider;

Example	Not Heavy	Smelly	Spotted	Smooth	Edible
A	1	0	0	0	1
B	1	0	1	0	1
C	0	1	0	1	1
D	0	0	0	1	0
E	1	1	1	0	0
F	1	0	1	1	0
G	1	0	0	1	0
H	0	1	0	0	0
U	0	1	1	1	?
V	1	1	0	1	?
W	1	1	0	0	?

You know whether or not mushroom A through H are poisonous, but you do not know about U through W. Which attributes should you choose as the root of a decision tree? Give reason.

[8]

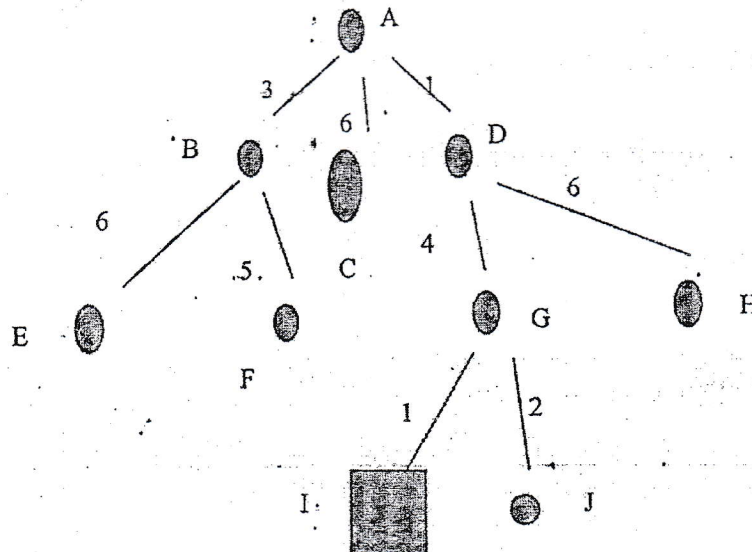
8. What are the major problems associated with NLP? Explain different steps involved in NLP. [2+6]
9. Why do you need Multilayer Perceptron Neural Networks? Write down the perceptron algorithm and use it to Construct Network which perform like Logical AND Operation and show each calculation performed. [2+6]
10. Write short notes on: [2×4]
- a) Uninformed search
 - b) Supervised Vs Unsupervised learning

Exam.	Regular / Back		
Level	BE	Full Marks	80
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Year / Part	IV / II	Time	3 hrs.

Subject: - Artificial Intelligence (Elective III) (CT 78506)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- Describe Artificial Intelligence and its subfields. What is Total Turing Test? [4+3]
- How do you define a problem in a state space? In a correctly worked out crypt-arithmetic problem $AB + CD = AAA$, what could be the possible values of B? Justify your answer. [3+4]
- Explain how alpha-beta pruning helps us reduce the complexity of adversarial search. Consider a search tree for a path-finding problem as shown. Assuming node A to be the root node and node I to be the goal node, explain how Best First Search reaches the goal by showing changes in the open list, closed list, and path at each step of the search. [4+5]



- Convert the following into FOL. [2+5]
 - Arya hates Joffery and Cersei.
 - No Lannister loves Bran.
 - Sansa has at least one sister.
 - Every Stark hates at least one Lannister.
 What are the steps involved in converting FOL into CNF?
- Consider the following axioms. [7]

All direwolves howl at night.
 Anyone who has any cat will not have any mice.
 Lights sleepers do not have anything that howls at night.
 Jon has either a cat or a direwolf.
 Prove that if Jon is a light sleeper, then Jon does not have any mice using resolution method.

6. What do you understand by structured knowledge representation? Explain semantic nets. [2+5]
7. Differentiate between supervised and unsupervised learning methods. Explain the steps of genetic algorithm with an example. [5+5]
8. Describe Expert System with its architecture. What are its advantages and disadvantages over a human expert? Explain any one expert system that you know of. [5+2+2]
9. Explain the steps of Natural Language Processing. Explain syntactic analysis and draw a parse tree for the following sentence. "A girl has a name." [5+4]
10. What are the issues present in knowledge representation in an expert system? A perceptron can learn a linearly separable function when given enough training. Justify with appropriate figures. [2+6]

Examination Control Division**2074 Magh****Exam.****Back****Level**

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Subject: - Artificial Intelligences (Elective III) (CT78506)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- ✓ Assume suitable data if necessary.

1. What is Artificial Intelligence, Knowledge and Learning? "Learning is an essential characteristic for intelligent agents". Comment on this statement. [2+2+2+4]
2. Discuss about Constraints Satisfaction Problem (CSP). Solve the following Crypt-arithmetic problem. [8]

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3. Make a detail comparison between Informed and Uniformed search with examples. [8]
4. Why Conjunctive normal form is required? Explain all the steps to convert to CNF. Transform " $P \rightarrow ((Q \ \& \ \sim R) \leftrightarrow S)$ " into CNF. [6]
5. Differentiate between Supervised and Unsupervised Learning. Clustering fall under which of the above categories? Justify stating algorithms. [2+4]
6. What are frames and semantic networks? Compare them with suitable examples. [2+5]
7. Differentiate between NLU and NLG. List down the different steps involved in the natural language processing (NLP) with suitable examples. [9]
8. What do you understand by Perceptron? How can we design a neural network that acts as an XOR gate? [10]
9. How best attribute is selected in a decision-tree? Select first best attribute of the decision-tree from given sample data. [8]

<u>Outlook</u>	<u>Temperature</u>	<u>Humidity</u>	<u>Windy</u>	<u>Play gold (Target variable)</u>
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

10. Write short notes on:

- i) Alpha-beta pruning
- ii) Self Organizing Map (SOM)

[2×4]

Examination Control Division
2074 Bhadra

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Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Artificial Intelligence (Elective III) (CT78506)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
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1. What are intelligent agents and how can we design intelligent agent? Explain with examples with relevance to PEAS framework? [4+3]

2. What do understand by Constraint satisfaction problem? Solve the following Crypt-arithmetic problem. [8]

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3. Compare breadth first search and depth first search along with examples. [8]
4. State the approach for learning using ID3 and Select the root attribute of the decision-tree from given sample data. [2+7]

Outlook	Temperature	Humidity	Windy	Play golf (Target variable)
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

5. Assume the following facts: [8]
- a) Horses, cows, pigs are mammals.
 - b) An offspring of a horse is a horse.
 - c) Bluebeard is a horse.
 - d) Bluebeard is Charlie's parent.
 - e) Offspring and parent are inverse relations.
 - f) Every mammal has a parent.
- Prove Charlie is a horse using resolution refutation
6. "Learning is an essential characteristic for intelligent agents." Comment on this statement. Differentiate between Supervised and Unsupervised Learning. [4+4]
7. Define a NLU and a NLG. Listdown the different steps involved in the natural language processing (NLP) with suitable examples. [2+6]
8. What do you understand by Perceptron? How can we design a neural network that acts as an XOR gate. [1+8]
9. What are frames and semantic networks? Compare them with suitable examples. [4+3]
10. Write short notes on the following: - [2*4]
- a) Minimax algorithm
 - b) Genetic algorithm

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

55
- 27
28

Subject: - Artificial Intelligence (Elective III) (CT78506)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
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1. What is Artificial Intelligence? What are the advantages and limitations of Artificial Intelligence over Natural Intelligence? [7]
2. Solve the following puzzle by assigning numeral (0-9) in such a way that each letter is assigned unique digit which satisfy the following addition.
BASE+BALL=GAMES [8]
3. Compare Greedy Search and A* search along with a simple example. [8]
4. Assume the following facts:
John likes all kinds of food.
Apples are food.
Chicken is food.
Anything anyone eats and isn't killed by is food.
Bill eats peanuts and is still alive.
Sue eats everything bill eats.
Prove that John likes peanuts using resolution. [8]
5. Why conjunctive normal form is required? Explain all the steps to convert to CNF. Transform following into CNF
a) $P \vee (\neg P \wedge Q \wedge R)$
b) Everyone who loves all animals is loved by someone. [2+2+2+3]
6. What is perceptron? How can we design a neural network that acts as an AND gate. [2+6]
7. Explain the different steps involved in the natural language processing (NLP) with suitable block diagram and examples. [7]
8. Explain in details four type of Knowledge Representation Scheme. [8]
9. In a County, 51% of the adults are males and the other 49% are females. One adult is randomly selected for a survey involving credit card usage.
Find the prior probability that the selected person is a male.
It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration).
Use this additional information to find the probability that the selected subject is a male. [8]
10. Compare the followings: [3x3]
a) Mutation versus Crossover
b) Induction versus Deduction
c) Forward versus Backward Chaining

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Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Artificial Intelligence (Elective III) (CT78506)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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1. Define an *Artificial Intelligence* stating the future prominence of the field. Which period was considered as '*AI research failure*' period? Why? Write in brief. [7]
2. Consider a Tic-Tac-Toe game playing as a problem. Prepare a brief description of your problem in terms of complete state-space, efficient data structures and goal, which are to be used for a machine to play that game. [7]
3. Though BFS and DFS have some definite way of making search, but still termed as blind search, why? Justify it comparing with any one informed search technique. [7]
4. Given the following premises: [6+4]

Bhogendra likes all kinds of food. Oranges are food. Chicken is food. Anything anyone eats and isn't killed by is food. If a person likes a food means that person has eaten it. Jogendra eats peanuts and is still alive. Shailendri eats everything Bhogendra eats.

~~Express~~ them in a FOPL, put them into the clause form, and use the resolution to prove that Shailendri likes chicken.
5. What is a semantic net? Draw a sample net about your course AI to demonstrate the knowledge representation features and qualities of semantic net. [7]
6. Explain the working principle of genetic algorithm with some illustrative examples. [7]
7. Explain a multi-layer perceptron working principle with clear statement of related theory. [7]
8. What are the different levels of analysis required in NLP? Justify with one example of each. [6]
9. Differentiate between declarative versus procedural knowledge processing in expert system. [6]
10. Write short notes on [4*4]
 - a) A* search
 - b) Sentence validity
 - c) Learning based on statistics
 - d) Learning rate

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Artificial Intelligence (Elective III) (CT78506)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. What is an Artificial Intelligence, Knowledge and Learning? "Learning is an essential characteristic for intelligent agents". Comment on this statement. [2+2+2+4]
2. Discuss about a Constraints Satisfaction problem (CSP). Solve the following Crypt-arithmetic problem. [8]

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3. Explain about depth first and best first search and compare in terms of space and time complexity. [8]
4. Why a Conjunctive normal form is required? Explain all the steps to covert to CNF. Transform " $P \rightarrow ((Q \text{ and } \sim R) \leftrightarrow S)$ " into CNF. [8]
5. After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e, the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given that you don't have the disease). The good news is that this is a rare disease, striking only one in 10,000 people. Why is it good news that the disease is rare? What are the chances that you actually have the disease? [8]
6. Differentiate between Supervised and Unsupervised Learning. Clustering fall under which of the above categories? Justify stating algorithms. [2+4]
7. How best attribute is selected in a decision-tree? Select first best attribute of the decision-tree from given sample data. [8]

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Rainy	Hot	High	True	No
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Overcast	Cool	Normal	True	Yes
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Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

8. What is the significance role of machine vision help in Artificial Intelligence? Explain how back propagation algorithm works. [3+5]
9. Define a NLU and a NLG? Explain the different ambiguities of Natural Language Processing. [1+7]
10. Write short notes on: [2×4]
 - a) Alpha-beta pruning
 - b) Self Organizing Map (SOM)



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TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

Examination Control Division

2071 Bhadra

Exam.	Regular / Back		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - Artificial Intelligence (*Elective III*) (CT78506)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
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1. What are main growth areas of Artificial intelligence? Describe with some implementations. Also compare and contrast between belief, hypothesis and knowledge. 4+4
2. a. You are given two jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4 gallon jug? Solve it by specifying all steps and properties of problem solving. 4
b. Define a production system. What are essential terms or steps of production system? 4
3. Describe Breadth First Search in detail. Also describe why Breadth First Search is Uninformed or blind search. 8
4. Define a WFF with its properties. Prove that preposition $(A \wedge (A \rightarrow B) \rightarrow B)$ is tautology. 4+4
5. Define resolution. You are given following statements. 8
 - a. Ram likes all kinds of food.
 - b. Apples are food.
 - c. Chicken is food.
 - d. Anything anyone eats and is not killed by is food.
 - e. ~~Krishna eats peanuts and is still alive.~~
 - f. Radha eats everything Krishna eats.Prove the statement 'Ram likes peanut' using resolution.
6. Define a frame. What is significance of frame in artificial intelligence? Explain with example. 8
7. What are the different ways of learning? Describe each in detail. 8
8. What are features of Hopfield network? How Hopfield network determine active and passive processing unit in network? Clarify with necessary example and illustration. 8
9. What is an expert system? Describe architecture of expert system with diagram. 8
10. Define following terms 8
 - a. Phonetic analysis
 - b. Syntactic analysis
 - c. Semantic analysis
 - d. Pragmatic analysis

Exam.	Regular	
Level	BE	Full Marks 80
Programme	BEI	Pass Marks 32
Year / Part	IV / I	Time 3 hrs.

Subject: - Artificial Intelligence (CT 710)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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1. How can you define AI from the dimension of behavioral process and thought process? Using your own assumptions, design PEAS framework for medicine delivery drones. [2+5]
2. In problem solving what is the concept of state space, state, successor function, goal test and path cost? Solve the following crypto arithmetic problem by showing all the steps.
B A S E + B A L L = G A M E S [3+5]
3. Define the terms admissibility and optimality in the context of A* search. Under what conditions is A* guaranteed to be both admissible and optimal? Provide justification for your answer. [4+5]
4. a) A Bayesian Belief Network (BBN) is used in a medical diagnosis system. The network includes the following binary variables: [7]
 - C: Patient has Cancer (Yes / No)
 - T: Test result is Positive (Yes / No)
 - S: Patient is a Smoker (Yes / No)
 - The network structure is: $S \rightarrow C \rightarrow T$
 - Conditional probabilities are given as:

$P(S=Yes)=0.3$	$P(C=Yes S=Yes)=0.2$	$P(C=Yes S=No)=0.05$
$P(T=Yes C=Yes)=0.9$	$P(T=Yes C=No)=0.1$	

Explain how the Bayesian Belief Network enables reasoning under uncertainty in this medical context.
- b) Using given statement prove "Mr. A is a criminal" [6]

It is crime for a Nepali to sell weapons to enemy of Nepal. Mr. A is an enemy of Nepal. Mr. A has some missiles. Missiles are weapon. Mr. R is a Nepali. If Mr. A has missiles then those missiles were sold to Mr. A are by Mr. R.
5. What are different issues in knowledge representation? Represent the following sentences using semantic net representation.
Tom is a cat. Tom caught a bird. Tom is owned by Sudeep. Tom is ginger in color. Cats like cream. The cat sat on the mat. A cat is a mammal. A bird is an animal. All mammals are animals. Mammals have fur. Sudeep age is 20. [2+5]
6. Define fuzzy logic and explain how it differs from classical binary logic? Explain Genetic Algorithm with suitable example. [2+6]
7. Design a Hopfield network to store three binary patterns of length 4. Show how to compute the weight matrix and demonstrate the network's ability to recall a pattern from a noisy input. [6]
8. How can ambiguity occur at phonetic, syntactic, semantic and pragmatic levels of natural language processing? Give an example. How syntactic and semantic analysis is done during natural language processing? [3+4]
9. Define Expert System. How does it simulate human reasoning? Explain with example. [2+5]
10. What do you understand about Perception? How can we design a neural network that acts as an XOR gate? [2+6]

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2082 Baishakh

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Artificial Intelligence (CT 710)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
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1. What is intelligent agent? How can you design an intelligent agent? List down PEAS for self-driving CAR. [2+6]
2. Discuss the steps involved in problem solving. Solve the following crypto-arithmetic problem by stating all the necessary constraints. [2+6]

$$\text{BASE} + \text{BALL} = \text{GAMES}$$

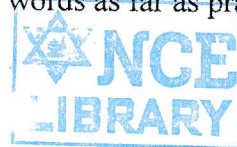
3. What are the drawbacks of greedy best first search method and explain how A* algorithm solves it with a suitable example? [2+6]
4. What is knowledge-based agent? How does it work? Explain with examples. [2+3+2]
5. What are casual networks? Given the following statistics, what is the probability that a woman has cancer if she has positive mammogram result? [2+6]
 - i) One percent of women over 50 have breast cancer.
 - ii) Ninety percent of women who have breast cancer test positive on mammograms.
 - iii) Eight percent of woman will have false positives.
6. When do we use semantic network? Convert the following sentences into semantic network. [2+6]
 - i) Employee is a human.
 - ii) Hari is an employee.
 - iii) Sita is a human.
 - iv) Supervisor is a human.
 - v) Ram is a supervisor.
 - vi) Ram is married to Sita.
 - vii) Ram is supervisor of Hari
7. When do we use fuzzy logic? Explain the architecture and working mechanism of fuzzy inference system. [2+6]
8. You are stuck in a disaster-prone area of Nepal. Give a specific example how you design expert system in this scenario with its block diagram. Highlight the key components of the designed system. [8]
9. Define a NLU and a NLG. List down the different steps involved in the natural language processing (NLP) with suitable examples. [2+6]
10. Write short notes on: [3×3]
 - a) Adversarial search
 - b) Hopfield network
 - c) Machine vision applications

TRIBHUVAN UNIVERSITY
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1. If the Turing test is passed, does this show that computer exhibit intelligence? Explain. Compare it with Reverse Turing test with an example. [4+3]
2. What do you understand by constraint satisfaction problem? Solve the following crypto-arithmetic problem by stating all the necessary constraints. [2+6]

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3. Compare informed and uninformed search. Explain how the Simulated Annealing algorithm to overcome the demerits of hill climbing search. [3+6]
4. Consider the following axioms: [9]
 - Every American who sells weapons to hostile nations is a criminal.
 - Every enemy of America is a hostile.
 - Nono has some missiles.
 - All missiles of Nono were sold by Gorge.
 - Gorge is an American. Nono is a Country.
 - Nono is the enemy of America.

Missiles are weapons. Prove that "Is Gorge a criminal?" using Resolution by Refutation.

5. Differentiate between forward chaining and backward chaining with examples. [7]
6. Explain different approaches of knowledge representation. Highlight pros and cons of Semantic Nets and Frames. Use them to represent following sentences. [2+6]
 - Figo is a dog.
 - Fido is own by Rohit
 - Fido is ginger in color.
 - Dog is a mammal.
 - All Mammals are animals.
 - Mammals have fur.

7. When do we use genetic Algorithm? Explain all steps in genetic algorithm with block diagram and operators. [2+6]
8. Define Perception. Justify the statement "Perceptron can work was Linear Classifier" taking reference of Logic gates. [2+6]
9. Define Machine Vision. Discuss the steps involved in Machine Vision with an example. [2+5]
10. Write short notes on: [3×3]
 - a) A* Search
 - b) Kohonen Network
 - c) NLU & NLG

TRIBHUVAN UNIVERSITY
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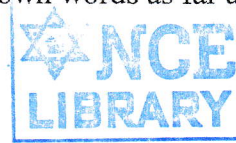
1. Define intelligent agent. Explain the factors that are required to pass the Turing Test. [2+6]
2. What is well-defined problem? Solve the following crypto arithmetic problem by showing all the steps. [2+6]
LOVE + LOVE = HATE
3. Compare the informed and uniformed search techniques. What strategies can be employed to address the issue of local maxima in Hill Climbing Search? [4+4]
4. Define skolemization with example. Consider the following axioms: [2+6]
 - a) Anyone passing his Engineering exams and winning the lottery is happy.
 - b) Anyone who studies or is lucky can pass all his exams.
 - c) Sneha did not study but she is lucky.
 - d) Anyone who is lucky wins the lottery.By using resolution refutation, prove that Sneha is happy?
5. What is prior probability and posterior probability? A doctor knows that the disease meningitis causes a patient to have a stiff neck, say, 80% of the time. The doctor also knows some unconditional facts: the prior probability that any patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1%. Determine the probability of meningitis when a patient has a stiff neck. [2+6]
6. Compare semantic net and frame. How you convert semantic net into frame? Explain with example. [3+5]
7. What are different methods for learning? Explain the steps involved in a genetic algorithm with a flowchart and justify why mutation is important. [3+5]
8. What is perception? Construct a Hebbian Network that performs like AND Gate. [2+6]
9. What is expert system? Explain how inference is performed in expert system? [3+5]
10. Write short notes on: [2×4]
 - a) Fuzzy Logic
 - b) Natural Language Processing

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Examination Control Division
2080 Bhadra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Artificial Intelligence (CT 710)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.



1. Define Artificial Intelligence and Intelligent agent. Differentiate between different types of intelligent agent with examples. [2+6]
2. What is constraint satisfaction problem? Solve the following cryptarithmic problem. [2+6]
TEN + TEN + FORTY = SIXTY
3. How searching is done in adversarial search? How A* search overcomes problem associated with Greedy Search? Explain with example. [2+6]
4. Consider the following axioms:
 - i) Ram likes all kinds of food.
 - ii) Apples and vegetables are food.
 - iii) Anything anyone eats and not killed is food.
 - iv) Anil eats peanuts and is still alive.
 - v) Ram eats everything that Anil eats.

Proof that "Ram likes peanuts" using resolution refutation. [8]
5. What do you understand by Bayesian Network? In a County, 51% of the adults are males and the other 49% are females. One adult is randomly selected for a survey involving credit card usage. [2+6]
 - i) Find the prior probability that the selected person is a male.
 - ii) It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration).

Use this additional information to find the probability that the selected subject is a male.
6. How a knowledge representation is evaluated? Explain semantic net and frame representation with examples. [3+5]
7. Explain all the steps in Fuzzy Learning with block diagram with an example. [8]
8. What are the problem associated with NLP? How syntactic and semantic analysis is done during natural language processing? Explain with example. [3+5]
9. Describe mathematical model of neural network. What does it means to train a neural network? Write an algorithm to its learning. [2+2+4]
10. Write short notes on: [2×4]
 - a) SOM
 - b) Expert Systems

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Examination Control Division
2080 Baishakh

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Artificial Intelligence (CT 710)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.



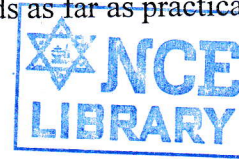
1. What is an intelligent agent? What type of model it works on? Explain PEAS for a Medical Diagnosis System. [2+2+3]
2. What do you understand by constraint satisfaction problem? Solve the following crypto-arithmetic problem. EAT+THAT = APPLE [2+6]
3. What are the problem associated with hill climbing search? Explain how simulated annealing overcome problem associated in hill climbing search. [3+5]
4. Assume the following facts:
 - a) Sita likes all kinds of food.
 - b) Momo and Burger are food.
 - c) Anything anyone eats and isn't killed by is food.
 - d) Bill eats mushroom and is still alive.
 - e) Sue eats everything Bill eats.Prove that "Sita likes mushroom" using resolution refutation method. [8]
5. What is Bayesian network? Describe with an appropriate example. How inference can be done using this network? [5+3]
6. What are the different knowledge representation models? Discuss about semantic nets with an example. [2+5]
7. Discuss the steps involved in genetic algorithm with example. [8]
8. What are the problem associated with NLP? How syntactic morphological and semantic analysis is done during natural language processing? Explain with example. [2+5]
9. What are major characteristics of expert system? Explain with example. Explain a general approaches for developing an expert system with a block diagram. [2+5]
10. Write short notes on: [3×4]
 - a) Fuzzy logic
 - b) Computer vision
 - c) A* Search.

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2079 Bhadra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Artificial Intelligence (CT 710)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.



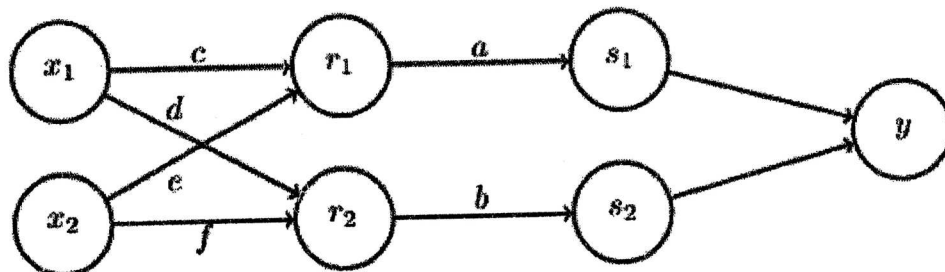
1. What is Turing test? How is can be used to measure intelligence of machine? Explain. [2+5]
2. What are the steps of Problem solving? Assume you are given three empty jugs; a 3 – gallon, a 5-gallon and a 9-gallon, and a pump can fill water only in 3- gallon jug. How you get exactly 7 gallons of water in the 9- gallon jug? Formalize the problems; write down production rules and draw search tree to solve this water Jug problem. [2+1+2+3]
3. Different between Depth first search and Breadth first search with their performance criteria. [7]
4. Prove that "Charlie is a mammal" with given propositions. [8]
 - a) Cows, pigs and horses are mammal
 - b) The child of horse is a horse
 - c) Bluebeard is a horse
 - d) Bluebeard is charlie's father
 - e) Child and father are inverse relation
 - f) Every mammal has a father
5. Differentiate between forward chaining and backward chaining using suitable example. [7]
6. List the issues need to be consider in Knowledge Representation techniques. Convert given sentences into Semantic Network. [2+6]
 - a) Tom is a cat
 - b) Tom fights with rat
 - c) Tom is owned by Ram
 - d) Tom is black in color.
 - e) Cats like milk
 - f) Rats like cheese
 - g) The cat sat on the bed
 - h) A cat is a mammal
 - i) A Rat is an animal
 - j) All mammals are animals
 - k) Every mammal gives birth a baby
7. What do you mean by membership of an element in a fuzzy set? Explain the steps involved in Fuzzy Learning with suitable example. [2+6]
8. Define Artificial Neural Network. How non-linearity can be modelled through this ANN? Explain with example. [1+7]
9. List down the challenges faced by NLP. Why do you think machine vision is important in AI? Explain with example. [2+5]
10. Write short notes on: [3×4]
 - a) Hill climbing algorithm
 - b) SOM
 - c) Expert system

Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define Artificial Intelligence. Design the PEAS information Automated Robot in a manufacturing plant. [2+3]
2. What are criteria for defining problem? Solve the following crypto-arithmetic algorithm:
BASE + BALL = GAMES [2+6]
3. What do you mean by Searching and what are the types of Search methods? What are the drawbacks of Min-Max algorithm and explain how Alpha-Beta pruning algorithm solves it with a suitable example? [3+6]
4. a) Use resolution solve following problem: The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove That Col. West is criminal. [8]
b) Define production system. Define and differentiate forward and backward chaining. [2+4]
5. Define term frame. Convert the following sentences into semantic network. Cat is a mammal. Cat has fur. Mammal has vertebra. Mammal is an animal. Bear is a mammal. Bear has fur. Whale is a mammal. Whale lives in water. Fish is an animal 10. Fish lives in water. [1+6]
6. What is fuzzy logic? Explain the steps involved in genetic algorithm with examples. [2+6]
7. Below is a neural network with weights a, b, c, d, e, and f. The input are x_1 and x_2 . The first hidden layer computes $r_1 = \max(c.x_1 + e.x_2, 0)$ and $r_2 = \max(d.x_1 + f.x_2, 0)$. The second hidden layer and third layer computes s_1 , s_2 and y using sigmoid activation function. Suppose the network has inputs $x_1 = -1$ and $x_2 = -1$. The weight values are $a=1$, $b=1$, $c=4$, $d=1$, $e=2$ and $f=2$. Perform forward propagation. And update weight for second hidden layer for a and b. [8]



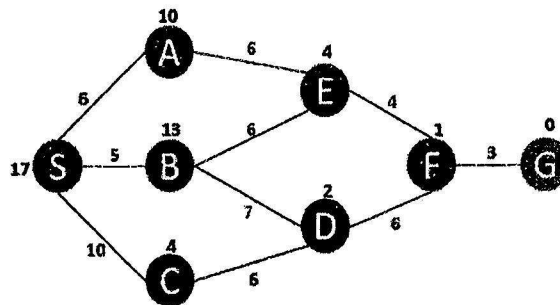
8. Define term expert system and expert system Shells. What are the characteristics that problem possesses are solved by expert system? Explain with example. [2+2+3]
9. Explain the McCulloch-Pitts model of a neuron with a diagram. Explain with example how to choose best activation function in particular project? [4+3]
10. What are the problem associated with NLP? List down the different steps involved in the natural language processing with suitable example. [2+5]

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- ✓ Assume suitable data if necessary.

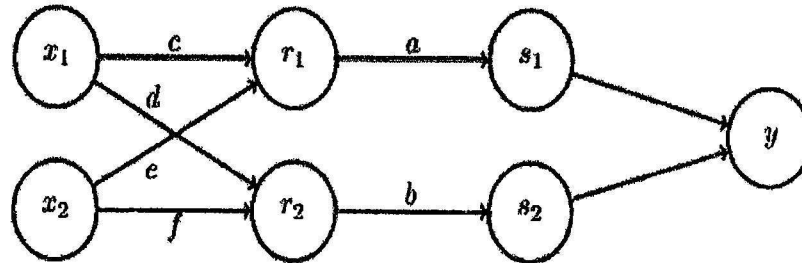
- How you define Artificial Intelligence (AI). Explain AI applications in different areas as problem solving tool. [2+4]
- Explain steps of problem solving with suitable example. Solve the following crypto-arithmetic algorithm SWIM + WEAR = RELAX. [3+6]
- Explain A* algorithm in following figure where starting node is S and destination is G. (Note: Information in the node represents cost required to reach the goal and information in the connecting link represents cost from one node to the other. [8]



- Assume the following facts:
Every child loves Santa. Everyone who loves Santa loves any reindeer. Rudolph is a reindeer, and Rudolph has a red nose. Anything which has a red nose is weird or is a clown. No reindeer is a clown. Scrooge does not love anything which is weird. Using Resolution Refutation method derive the following conclusion: Scrooge is not a child. [8]
 - What is CNF? Explain rule of inference along with example. [2+3]
- Define term semantic net. Represent the following sentences using semantic net representation.
Tom is a cat. Tom caught a bird. Tom is owned by Sudeep. Tom is ginger in color. Cats like cream. The cat sat on the mat. A cat is a mammal. A bird is an animal. All mammals are animals. Mammals have fur. Sudeep age is 20. [2+6]
- Name and describe the main features of Genetic Algorithms along with example. [7]
- Define Expert System with example. Explain the limitations of Expert System. [4+3]
- What is the role of activation function in ANN? Explain the McCulloch-Pitts model of a neuron with a diagram. [3+4]

9. Below is a neural network with weights $a, b, c, d, e,$ and f . The input are x_1 and x_2 . The first hidden layer computes $r_1 = \max(c.x_1 + e.x_2, 0)$ and $r_2 = \max(d.x_1 + f.x_2, 0)$. The second hidden layer and third layer computes s_1, s_2 and y using sigmoid activation function. Suppose the network has inputs $x_1 = -1$ and $x_2 = -1$. The weight values are $a = 1, b = 1, c = 4, d = 1, e = 2$ and $f = 2$. Perform forward propagation. And update weight for second hidden layer for a and b .

[8]



10. What are the problem associated with NLP? List down the different steps involved in the natural language processing with suitable example

[2+5]

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2081 Ashwin

Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is intelligent agent? Do we need PEAS for designing an intelligent agent? If so, define PEAS and provide the PEAS information for Taxi Driver Agent and Automated Robot in a manufacturing plant. [1+5]
2. List down the characteristics of well-defined problem. [8]

You are given two jugs, a 5-litre one and a 2-litre one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. You are required to measure exactly 1 liter of water. Is this well-defined problem? Justify your argument and list all the attributes to design an intelligent agent. Solve this problem by specifying states and the operations used.
3. List down the approaches for evaluating searching methods. What are the drawbacks of Min-Max algorithm and explain how Alpha-Beta pruning algorithm solves it with a suitable example? [2+6]
4. What do you mean by Forward and Backward Chaining? The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove That Col. West is criminal using forward chaining. [2+6]
5. What are the approaches for measuring uncertainty? [2+6]

1% of people have a certain genetic defect. 90% of tests for the gene detect the defect (true positives). 10% of the tests are false positives.

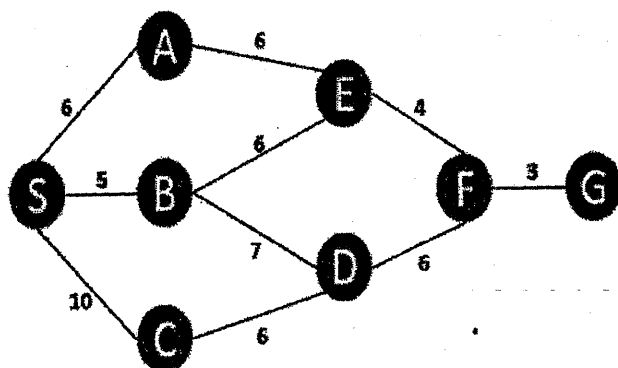
 - a) What might be the probability of getting the genetic defect result?
 - b) If a person gets a positive test result, what are the odds they actually don't have the genetic defect?
6. What are Frames and Semantic Network? Convert the following sentences into FOPL and represent using semantic network. [2+6]
 - i). Ram is a person. (ii) Person is an animal. (iii) Ram likes Jhilke.
 - iv) Jhilke is a cat. (v) Cat eats fish. (vi) Cat is an animal.
 - vii) fish is an animal.
7. When shall we use genetic algorithm? Explain with appropriate example. Discuss about the steps in genetic algorithm including selection, crossover and mutation operations. [3+5]
8. You are hired by a company to work on product based on a Nepali Language. What are challenges/issues associated with NLP? List them with appropriate examples and relate these with Nepali Language. What are your plans to mitigate these challenges while working in the company? [8]
9. What is Machine Vision? List down the steps involved in Machine Vision by giving suitable example. [2+4]
10. Write Short notes on: [3×4]
 - a) Fuzzy Logic (b) Learning in Neural Network (c) Expert System

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- Can machine think? Explain your stance with appropriate reasons. Relate it with Turing Test and Reverse Turing Test. [2+4]
- What do mean by Constraints Satisfaction Problem? List all the constraints and solve the following crypto-arithmetic problem: EAT + THAT = APPLE [2+5]
- How A* search overcomes problem associated with Greedy Best First Search? Using following figure and table where starting node is S and destination is G, compare the result of A* algorithm with greedy search. [3+4+3]



From	Cost
S	15
A	10
B	12
C	5
D	4
E	2
F	1
G	0

(Note: Table gives the cost required to reach the goal and information in the connecting link represents cost from one node to the other.)

- List down the steps for converting to CNF and proof by resolution refutation. Convert the following sentences into FOPL and answer "Did Curiosity kill the cat" by resolution refutation. [3+5]
 - Everyone who loves all animals is loved by someone.
 - Anyone who kills an animal is loved by no one.
 - Jack loves all animals.
 - Either Jack or Curiosity killed the cat.
- Discuss about the different approaches knowledge representation. How do you make choice among these approaches? Is it possible to evaluate the knowledge representation? List down any issues associated with it. [2+2+2+2]
- When shall we use Fuzzy Logic? Explain with appropriate example. What are the steps of major steps for developing system based on Fuzzy logic? [2+5]
- What is Prior Probability and Posterior Probability and show how these probabilities are used in Bayes Theorem. Is there any connection between Belief Network and Bayes Theorem? Justify with an example. [3+5]
- You are hired by a company to develop a Nepali Chatbot System. What are the knowledge bases that is required to develop the system? List down the different steps involved in the development of this Nepali Language based NLP tool with suitable block diagram and examples. [2+6]

9. You need to work as a Knowledge Engineer for a company in developing expert system for Disease Diagnosis. Draw a block diagram and highlight the major steps for development of this expert system.

[6]

10. Write Short notes on:

[3×4]

- a) Forward chaining and Backward chaining
- b) Hopfield Neural Network
- c) Machine Vision

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Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

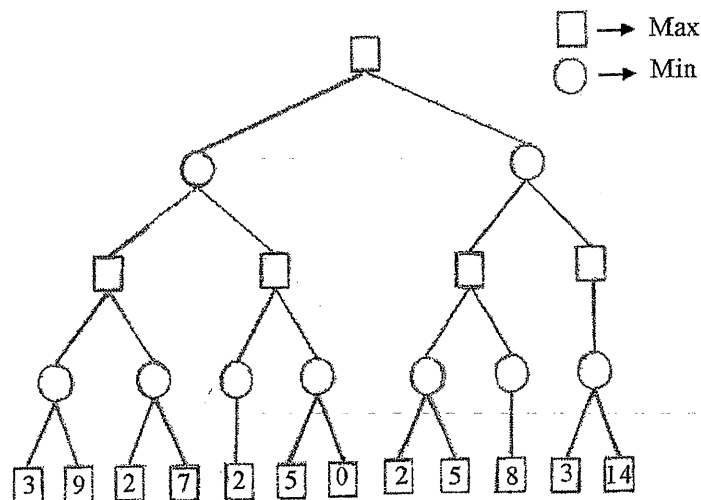
1. What is Artificial Intelligence? Explain different types of Intelligent Agents with examples. [2+6]
2. Define term well defined problem. Solve the following crypto arithmetic problem by showing all the steps. [2+6]
ONE + ONE + TWO = FOUR
3. Why is searching important in problem solving? Differentiate between Depth first search and Breadth first search with their performance criteria. [2+6]
4. What are different issues in knowledge representation? Discuss the importance of frame in knowledge representation with suitable example. [3+5]
5. List the rules of inference. Convert the following into FOPL and prove using Resolution Refutation System. "All people who are not poor and are smart are happy. Those people who can read are not stupid. John can read and is wealthy. Happy people have exciting lives. Can anyone be found with an exciting life? [3+5]
6. When do we need hopfield Neural Network? Make a comparision between feed forward network and hopfield Network. [3+5]
7. How reasoning is done in uncertainty? Three factories produce light bulbs to supply the market. Factory A produces 20%, 50% of the tools are produced in factories B and 30% in factory C. 2% of the bulbs produced in factory A, 5% of the bulbs produced in factory B and 3% of the bulbs produced in factory C are defective. A bulb is selected at random in the market and found to be defective. What is the probability that this bulb was produced by factory B? [3+5]
8. Define Machine learning. What is Fuzzy logic? Explain Fuzzy inference with suitable example. [1+2+5]
9. Compare expert systems and human experts. Explain each point with suitable practical examples. [8]
10. What is machine vision? Create a senario where we require Machine vision. List down the steps how you solve the above problem. [2+6]

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Define Artificial Intelligence (AI)? Differentiate between Natural (Human) Intelligence and Artificial Intelligence. Can AI replace Humans? Justify. [1+3+4]
2. What is Constraint Satisfaction problem? Solve the following problem LOGIC + LOGIC = PROLOG. [3+5]
3. What is minmax search? Apply minmax search along with alpha-beta pruning for the following tree at max node at top. [2+6]



4. What is semantic network? Convert the following sentences into semantic network.
 - i) The avg height of an adult is 5'6".
 - ii) Cricket player is an adult male.
 - iii) Adult male is a person.
 - iv) Batting average of a cricket player is 37.59
 - v) Rohit Paudel is a Fielder.
 - vi) Fielder is a cricket player.
 - vii) Team of Rohit Paudel is Nepal.
5. Why CNF is necessary? "Everyone who loves all animals are loved by someone" represent this statement in FOPL and explain all the steps involved to convert it into CNF. [2+6]
6. What is belief network? How reasoning is done in this network? [2+6]
7. What is evolutionary computing? List down the steps in genetic algorithm with an example. [2+6]
8. What is perceptron? When do we need back propagation algorithm? Explain with an example. [2+6]
9. What is expert system? Create a scenario where we require expert system. List down the steps how you create this system. [2+6]
10. When do we need Natural Language Processing? Explain different steps of NLP. [3+5]

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2079 Ashwin

Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is knowledge and learning? Why they are important? Explain the factors that are require to pass the Turing test. [2+2+4]
2. What is constraint satisfaction problem? Solve the following crypto-arithmetic algorithm:
BASE + BALL = GAMES. [2+6]
3. On what basis can we evaluate the search algorithms? Explain. What are the approaches for Depth First Search and Depth Limit Search? Explain. [4+4]
4. Assume the following facts:
 - a) Steve only likes easy courses.
 - b) Science courses are hard.
 - c) All the courses in basket weaving department are easy.
 - d) BK301 is a basket weaving course.

Prove that Steve likes BK301 course using Resolution Refutation Method. [7]
5. Why probabilistic reasoning is important in the AI? When do we use Bayesian Network? Explain. [3+6]
6. What are the different approaches for knowledge representation? List down the essential properties and issues of the knowledge representation system. [4+2+2]
7. What is Fuzzy logic? When do we use it? Explain the steps with suitable example. [1+2+5]
8. What is perceptron? Is single perceptron sufficient to realize a network behaving as XOR gate? Justify. [1+8]
9. Explain different steps involved in NLP. Why NLP is hard? Explain. [5+2]
10. What is machine vision? What are the different steps involved? List down its applications. [1+5+2]

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2079 Jestha

Exam. Level Programme Year / Part	Back		
	BE	Full Marks	80
	BCT	Pass Marks	32
	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define intelligent agent. Explain different types of intelligent agents and their interaction with environments along with appropriate examples. [1+6]
2. Discuss about Constraints Satisfaction Problem (CSP). Solve the following Crypt-arithmetic problem. [1+6]

$$\begin{array}{r} \text{BASE} \\ + \text{BALL} \\ \hline \text{GAMES} \end{array}$$

3. Differentiate between depth first search and breadth first search with an example. [7]
4. a) Assume the following facts: [6]
 - (i) Horses, cows, pigs are mammals.
 - (ii) An offspring of a horse is a horse.
 - (iii) Bluebeard is a horse.
 - (iv) Bluebeard is Charlie's parent.
 - (v) Offspring and parent are inverse relations.
 - (vi) Every mammal has a parent.

Prove Charlie is a horse using resolution refutation.
- b) Why Conjunctive normal form is required? Explain all the steps to covert to CNF. Transform each of the following sentences into CNF. [2+2+4]
 - (i) $\sim(P \ \& \ Q) \vee (P \vee S) \rightarrow R$
 - (ii) $P \rightarrow ((Q \ \& \ \sim R) \leftrightarrow S)$
5. What is perceptron? Can we design a neural network that acts as logic gates? Explain with an example. [1+6]
6. Differentiate between NLU and NLG. List down the different steps involved in the natural language processing (NLP) with suitable examples. [1+6]
7. Write short notes on the following: [3×5]
 - a) A* Search
 - b) Semantic Network and Frame
 - c) Self Organizing Map (SOM)

8. a) "Learning is an essential characteristic for intelligent agents." Comment on this statement. Differentiate between Supervised and Unsupervised Learning. [4+4]
- b) How best attribute is selected in a decision-tree? Select the root attribute of the decision-tree from given sample data. [1+7]

Outlook	Temperature	Humidity	Windy	Play cricket (Target variable)
Rainy	Hot	High	False	Yes
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	True	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	No
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

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Examination Control Division
2078 Chaitra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What are intelligent agents and how can we design intelligent agent? Explain with examples on relevance to PEAS framework. [4+4]
2. What is well defined problem? Solve the following crypto-arithmetic problem by defining it. [2+6]

TWO + TWO = FOUR
3. Why is searching important in problem solving? What are the drawbacks of greedy best – first search and how A* search technique is used to solve it. Explain with example. [2+7]
4. Define horn clause with example. Consider the following axioms:
 - i) Every child love Santa.
 - ii) Everyone who loves Santa loves any reindeer.
 - iii) Rudolph is a reindeer, and Rudolph has a red nose.
 - iv) Anything which has a red nose is weird or is a clown.
 - v) No reindeer is a clown.
 - vi) Scrooge does not love anything which is weird.

Represent these axioms in a predicate calculus and then convert each formula to CNF.

List down the rules of inference. When can we use these rules? Explain. [3+3]
5. Explain how statistical reasoning aids in inference and reasoning in light of Bayes theorem. At a certain University, 5% of men are over 6 feet tall and 2% of women are 6 feet tall. 60% of students are female. If a student is selected at a random from among all those over six feet tall, what is the probability that the selected students is woman? [3 +5]
6. Explain frame and Semantic Net with example. List down their advantages and disadvantages. [4+4]
7. List down the algorithms inspired by biological phenomena and social behaviors. Explain the learning process in Genetic Algorithm with suitable block diagram. [2+6]
8. Using Hebbian learning algorithm construct a neural network that behaves as an AND gate. Is hebbian learning supervised method? Justify. [6+3]
9. What are expert systems? Explain the architecture of expert system with suitable block diagram. Highlight the advantages and disadvantages of expert system. [1+4+3]
10. What is natural language understanding and natural language generation? Briefly list down the steps. Discuss the issues related with NLP. [2+4+2]

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2078 Poush

Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Compare human intelligence and machine intelligence. Discuss some applications of Artificial Intelligence. [3+5]
2. Discuss about Constraints Satisfaction Problem (CSP). Solve the following Crypt-arithmetic problem. [2+6]

FORTY + TEN + TEN = SIXTY

3. Justify how A* algorithm provides optimum solution compared with greedy search algorithm. [8]
4. Explain forward chaining giving suitable example. [8]
5. Assume the following facts: [8]
 - a. Horses, cows, pigs are mammals.
 - b. An offspring of a horse is a horse.
 - c. Bluebeard is a horse.
 - d. Bluebeard is Charlie's parent.
 - e. Offspring and parent are inverse relations.
 - f. Every mammal has a parent

Prove Charlie is a horse using resolution refutation

6. What are Frames and Semantic Net? Convert the given sentences in semantic Net: [2+6]
 - A person is a mammal.
 - Sandeep Lamichane is a person.
 - Person has nose.
 - Sandeep Lamichane is in Nepalese cricket team.
 - Uniform color of Sakti Gauchan is Red/Blue.
7. What is Machine Learning? Discuss about Supervised, Unsupervised and Reinforcement Learning with appropriate examples. [2+6]
8. What are problems of natural language understanding? Discuss the steps of NLP. [3+5]
9. Define Perceptron and multilayered perceptron. Design a neural network which acts as AND gate. [2+2+4]
10. Write short notes on the following: [2×4]
 - a) Adversial Search
 - b) Fuzzy logic

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INSTITUTE OF ENGINEERING
Examination Control Division
2078 Baishakh

Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

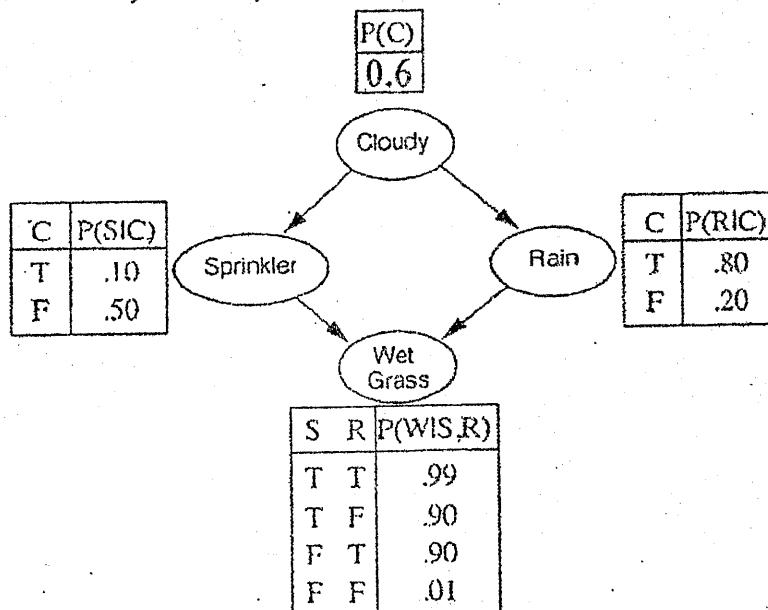
- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

- List down disadvantages of AI? What is the importance of the Turing Test in Artificial Intelligence? What are the applications of AI? Explain. [2+2+4]
- What do you understand by Constraint satisfaction problem? Solve the given Crypt-arithmetic problem: [2+6]

$$\begin{array}{r}
 \text{TEN} \\
 \text{TEN} \\
 + \text{FORTY} \\
 \hline
 \text{SIXTY}
 \end{array}$$

- Explain A* search method using a suitable example. Discuss the drawbacks of Hill Climbing Algorithm. [6+2]
- Assume the following facts:
 - Cow, Buffalo, Bull are mammals.
 - An offspring of a Cow is Cow.
 - Kali is a Cow.
 - Kali is Taarey's parent.
 - Offspring and parent are inverse relations.
 - Every mammel has a parent.
 By using resolution refutation method, prove that Taarey is a Cow. [8]

- What do you understand by Bayesian Network? For the given Bayesian network below, find the probability of grass being wet when the weather is cloudy, there is some sprinkler but no rain i.e. C = True, S = True, R = False and W = True. [2+6]



6. Differentiate between frames and semantic nets with an example. Why semantics nets and frames are important in AI? [6+2]
7. What is Machine Learning? Explain learning framework with suitable block diagram. [2+6]
8. What is perception? Construct a Hebbian network that performs like an AND gate. [1+7]
9. Explain the ambiguity in NLP. Discuss the different steps involved during NLP. [2+6]
10. Write short notes on: [2×4]
 - a) Supervised vs Unsupervised Learning
 - b) Min max algorithm

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2077 Chaitra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define Artificial Intelligence (AI). When is a machine said to have passed the Turing Test? Discuss two fields of application of AI. [2+3+3]
2. What do you understand by Production system problem? Solve the following crypto arithmetic problem. [2+6]

WRONG + WRONG = RIGHT

3. List out the disadvantages of MIN-MAX algorithm for Game playing and explain how Alpha-beta pruning helps to overcome the limitation of MIN-MAX algorithm with an example. [2+6]
4. It is a crime for an American to sell weapons to hostile nations. Nono has some missiles. All the missiles owned by Nono were sold to it by Colonel West. Missiles are weapons. An enemy of America counts as hostile. Colonel West is an American. The country Nono, is an enemy of America. Prove that Colonel West is a criminal by using FOPL based Resolution Refutation Method. [8]
5. What is rule-based reasoning? Explain with an example. How Bayes theorem wedin belief network? [2+6]
6. What do you mean by Conceptual Dependency? Explain how knowledge is represented using scripts. [3+5]
7. Explain all the steps in the genetic algorithm with block diagram and operators. [8]
8. What is neural network? Show that a single neuron cannot implement XOR gate. [2+6]
9. Draw the block diagram of an expert system and briefly explain each component. List the benefits of using expert system? [6+2]
10. Write short notes on: [2×4]
 - a) Hill climbing problems
 - b) Fuzzy learning

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2076 Baisakh

Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

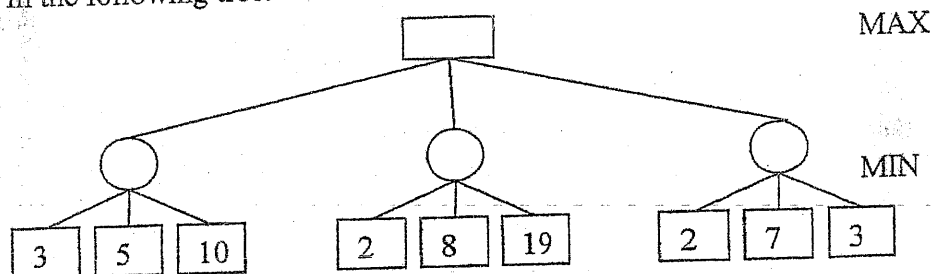
- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define Artificial Intelligence. Justify that "system that think rationally and act rationally." is part of Artificial Intelligence. [8]
2. Solve following crypto-arithmetic problem.

SEND+MORE=MONEY.

Assign different decimal digit to different letters. Explain the steps followed for the solution. [8]

3. Discuss about alpha-beta pruning algorithm. Find the value of min max value using this concept in the following tree. [4+4]



4. List down the rule for Inference. Consider the following axioms.
All hounds howl at night.
Anyone who has any cats will not have any mice.
Light sleepers do not have anything which howls at night.
John has either a cat or a hound.
Prove: "If John is a light sleeper, then John does not have any mice." By using resolution refutation. [2+8]
5. Define a semantic network and frames with an example. List advantages and limitations of both. [6+2]
6. What is machine learning? Explain learning by analogy with example. [2+6]
7. What is McCulloch/Pitts neuron? Can this neuron be trained, to represent EX-OR gate? It Justify and propose neural network model. [10]
8. What is an expert system? Explain the components of an expert system. [1+7]
9. Write short notes on: [3x4]
 - a) Boltzman Machine
 - b) Conjunctive Normal Form
 - c) A* Algorithm

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is an intelligent agent? How does learning agent work? [8]
2. What do you understand about well defined problems? Explain about problems that can be solved using production rules with an example. [2+6]
3. Discuss about the evaluation criteria for search algorithm. State the problems in hill climbing search algorithm. [4+4]
4. Why CNF is necessary? "Everyone who loves all animals are loved by someone" represent this statement in FOPL and explain all the steps involved to convert it into CFN. [2+6]
5. What is knowledge representation? How semantic network is used to represent knowledge? [2+6]
6. What do you understand by swarm intelligence? Suppose chromosomes are of the form $x = a b c d e f g h$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as: [2+8]
$$f(x) = (a+b) - (c+d) + (e+f) - (g+h)$$
 and let the initial population consist of four individuals with the following chromosomes.
7. What is Natural Language Processing (NLP)? Discuss the different steps in NLP with suitable examples. Also list down major issues in NLP. [6+2+2]
8. Explain Hopfield network with an example. [8]
9. Write short notes on: [3×4]
 - i) Predicate logic
 - ii) Unsupervised learning
 - iii) Breadth first vs depth first search

Exam.	Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Define AI. What is the importance of Turing Test in AI? List applications of AI. [2+4+2]
2. What is problem space? Solve the following crypto arithmetic problem by showing all the steps. [2+6]
$$\text{BASE} + \text{BALL} = \text{GAMES}$$
3. Discuss the hill climbing search algorithm along with problem associated with it and discuss their solutions. Why simulated annealing is important? [6+2]
4. Given premises "Every American who sells weapons to hostile nations is a criminal. The country XYZ is enemy of America. All of its misallies in XYZ were sold by Donald, who is an American." Prove that Donald is a criminal by using FOPL based resolution refutation method. [8]
5. Why CNF is required? Explain all the steps used to convert a quantified statement with suitable example. [2+6]
6. Why semantic network and frames are important in AI? Provide examples of both with FOPL statements example. [2+6]
7. What is a genetic algorithm? Explain all steps in genetic algorithm with block diagram and operators. [8]
8. List the importance of expert system in real life. Draw block diagram of expert system architecture and explain all blocks. [2+6]
9. What is a McCulloch/Pitts neural network? Explain it with reference to AND gate. Justify that it cannot be applied to Exclusive OR gate. [8]
10. Justify that NLP is one of the important part of an AI. Explain the steps involved in the NLP. [8]

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is a rational agent? "System that think like humans" and "System that act like humans" are the part of artificial intelligence. Justify these statement with practical example. [1+6]
2. Define constraint satisfaction problem (CSP). Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. SWIM + WEAR = RELAX. [1+6]
3. Why Searching is important in problem solving? What do the drawbacks of greedy best-first search and how A* search algorithm solve it. Explain with an example. [2+7]
4. a) Assume the following facts: [8]
 - John likes all kinds of food.
 - Apples are food.
 - Chicken is food.
 - Anything anyone eats and isn't killed by is food.
 - Bill eats peanuts and is still alive.
 - Sue eats everything bill eats.Prove that John likes peanuts using resolution.
- b) Differentiate between forward and backward chaining. [4]
5. What is Frame? How is it different from semantic net in knowledge representation? [7]
6. Define inductive Learning. Explain in detail about ID3 process with suitable example. [2+6]
7. What is self-organizing Map (SOM)? Explain all the steps involved in SOM with suitable example. [2+5]
8. Justify that the study of gene is one of important part in the AI. List down the steps involved in genetic algorithm with an example. [4+4]
9. How knowledge acquisition is performed in expert system? Explain one real expert system example with proper architecture. [2+5]
10. Write short notes on: (Any two) [2×4]
 - a) Machine Vision
 - b) Supervised Vs Unsupervised learning
 - c) Back Propagation Algorithm

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define Artificial Intelligence (AI). Describe the importance and practical application of AI. [6]
2. A farmer has a goat, a wolf and a cabbage on the west side of a river. He wants to get all of his animals and his cabbage across the river onto the east side. The farmer has a row boat but he only has enough room for himself and one other thing. The wolf will eat the goat if they are left together alone. The goat will eat the cabbage if they are left together alone. How can the farmer get everything on the east side? [8]
 - i) Formulate this puzzle as search
 - ii) Solve this problem-using search (any method)
 Draw the search tree and show the final solution
3. Devise an example to show how A* algorithm uses path cost and heuristic cost to generate best solution. [8]
4. Consider the following axioms: [10]
 - i) Anyone whom Mary loves is a football star
 - ii) Any student who does not pass does not play
 - iii) John is a student
 - iv) Any student who does not study does not pass
 - v) Anyone who does not play is not a football star.
 Prove that "If John does not study, then Mary does not love John" Resolution by Refutation.
5. A doctor is called to see a sick child. The doctor has prior information that 90% of sick children in that neighborhood have the flu, while the other 10% are sick with measles. Let F stand for an event of a child being sick with Flu and M stand for an event of a child being sick with measles. Assume for simplicity that there no other maladies in that neighborhood. A well-known (and common) symptom of measles is a rash and has probability of 0.95. [8]

However, very occasionally, children with flu also develop rash and has probability of 0.08.

Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?
6. Explain Frames and Semantic Net with examples. List down their advantages and limitations. [4+4]
7. What is Fuzzy Logic and why is it important? Explain about Mamdani Fuzzy Inference Method with example. [3+7]
8. What do you understand by Perception? How can we design a neural network that acts as an XOR gate? [1+7]
9. Differentiate between declarative knowledge and procedural knowledge. Describe expert system with its architecture and practical uses. [3+5]
10. What s Natural Language processing? Explain the different steps in the natural language processing. [2+4]

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. If the Turing Test is passed, does this show that computers exhibit intelligence? State your reasons. [7]
2. Solve the following puzzle by assigning numeral (0-9) in such a way that each letter is assigned unique digit which satisfy the following addition. [7]

$$\text{ONE} + \text{ONE} + \text{TWO} = \text{FOUR}$$
3. Explain the necessity of searching techniques in AI? Differentiate between Breadth first search and Depth first search with their performance criteria. [4+5]
4. Assume the following facts: [8]
 - i) Horses, cows, pigs are mammals
 - ii) An offspring of a horse is a horse
 - iii) Bluebeard is a horse
 - iv) Bluebeard is Charlie's parent
 - v) Offspring and parent are inverse relations
 - vi) Every mammal has a parent

Prove Charlie is a horse using resolution refutation.
5. What is causal net? How does Bayes Theorem calculate the probability in a causal net? Explain with example calculation. [7]
6. Convert given sentences into Semantic Network. [7]
 - i) The height of the adult male is 5.10
 - ii) Baseball player is an adult male.
 - iii) Adult male is a person.
 - iv) Batting average of Baseball players is 0.252
 - v) Pee-wee-Reese is a Fielder.
 - vi) Fielder is Baseball player.
 - vii) Team of pee-wee-Reese is Brooklyn Dodger.
7. "Learning is an essential characteristic for intelligent agents." List down justification on this statement. Write about the role learning with suitable example. [4+4]
8. What are applications of Expert System? Describe the Development stages of Expert System briefly. [2+6]
9. Define a NLU and a NLG. List down the different steps involved in the natural language processing (NLP) with suitable examples. [2+7]
10. Define Hebbian learning. Use Hebbian learning algorithm to Construct Hebbian Network which perform line AND Function. [3+7]

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. If the Turing Test is passed, does this show that computers exhibit intelligence? State your reasons. [7]
2. What is a Constraint satisfaction problem? Solve the following crypto-arithmetic problem with necessary steps:
LOGIC+LOGIC=PROLOG. [3+4]
3. Searching is an important part of AI, justify it. Explain any two types of blind search and compare them in terms of space and time complexity. [2+7]
4. What is a rule based reasoning? Explain Backward Chaining with suitable example. [7]
5. Using resolution solve the following statements: [7]
All pompeian are 'Romans. All Romans were either loyal to Caesor or hated him.
Everyone is loyal to someone. People only try to assassinate rulers they not loyal to.
Marcus tried to assassinate Caesor. Marcus was pompeian. Find, did Marcus Caesor?
6. What are Frames and Semantic Net? Convert the given sentences in semantic Net. [7]
 - i) A person is a mammal.
 - ii) Sakti Gauchan is a person.
 - iii) Person has nose.
 - iv) Sakti Gauchan is in Nepalese team.
 - v) Uniform color of sakti Gauchan is Red/Blue.
7. What is a machine learning? Explain genetic algorithm (GA) along with GA operators. List some of areas where GA can be applied. [2+6+2]
8. What do you understand by a perceptron? How can we design a neural network that acts as an OR gate. [2+7]
9. Differentiate between declarative knowledge and procedural knowledge. Describe expert system with its architecture and practical uses. [3+6]
10. Explain different steps involved in the natural language processing (NIP) with suitable block diagram and examples. [8]

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is an Artificial Intelligence (AI)? Explain any two applications of AI in real field. [7]
2. What do you understand by Constraint satisfaction problem? Solve the following Crypt-arithmetic problem. [1+6]

SEND
+ MORE
MONEY

3. What is a searching? Explain Breadth First Search and Depth First Search and compare their performance criteria. [9]
4. What is a knowledge, representation and reasoning? Describe forward chaining with practical example. [2+5]
5. Assume the following facts: [7]
 - John likes all kinds of food.
 - Apples are food.
 - Chicken is food.
 - Anything anyone eats and isn't killed by is food.
 - Bill eats peanuts and is still alive.
 - Sue eats everything Bill eats.

Prove that John likes peanuts using resolution refutation

6. What are semantic nets and frames? How frames are useful in semantic nets. [7]
7. What is a machine learning? Explain in detail about Boltzmann machines with suitable algorithm and explanations. [2+8]
8. What is a neural network? Explain the back propagation algorithms and perceptron. [2+4+4]
9. What is an expert system? Explain its advantages and disadvantages. [8]
10. What is a Natural Language Processing? Describe Natural Language Processing Steps and its application. [2+6]

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Distinguish between knowledge and learning. What does acting humanly refer to? Explain. Define well defined problems. [2+5+1]
2. Solve the following cryptoarithmetic problem with necessary steps. [7]
ONE + ONE + TWO = FOUR
3. How informed search is different from uniformed search? Explain min-max algorithm with suitable example, also discuss how alpha-beta is different from min-max algorithm. [2+4+3]
4. What is casual network? Explain reasoning in belief network with suitable example. [2+5]
5. What is neural network? Describe its types. A doctor knows that the disease meningitis causes the patient to have a stiff neck 50% of the time. The doctor also knows that the probability that a patient has meningitis is 1/50,000, and the probability that any patient has a stiff neck is 1/20. Find the probability that a patient with a stiff neck has meningitis? [3+5]
6. What is machine learning? Explain genetic algorithm along with different operators of genetic algorithm. [2+8]
7. What is Natural Language Processing (NLP)? Explain the syntactic, semantic and pragmatic analysis of NLP. [2+2+2+2]
8. Define expert system? Discuss about the general architecture of an expert system. [2+4]
9. How does machine vision help in Artificial Intelligence? Explain how back propagation algorithm helps in machine learning. [3+5]
10. Compare the followings: [3×3]
 - a) Forward versus backward chaining
 - b) Hopfield versus Kohonen network
 - c) Analogy versus Inductive learning

Exam.	Regular / Back		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is Artificial Intelligence (AI)? Discuss history of AI in brief. [2+5]
2. What is the advantage of depth limit search? Compare it with other search strategies. [4+4]
3. If X is on the top of Y, Y supports X. If X is above Y and they are touching each other, X is on the top of Y. A cup is above a book. A cup is touching a book. Show that supports (book, cup) is true. [8]
4. Give an example of learning by analogy. How knowledge can be represented using semantic network? Explain with suitable example. [1+7]
5. Explain Backward Chaining with suitable example. [7]
6. What is conceptual dependency? Explain some of common primitives used in conceptual dependency. [2+5]
7. What makes a problem "Well Defined"? Explain with a sample example of a state-space search framework. [3+4]
8. What is Artificial Neural Network (ANN)? Compare ANN with human brain with its functioning principle. [3+5]
9. Define knowledge acquisition with example. Explain the architecture of expert system. [2+6]
10. Compare the following: [4×3]
 - a) Declarative versus Procedural Knowledge
 - b) Pragmatic versus Phonetic Analysis
 - c) Genetic algorithm based versus Fuzzy based learning

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
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Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is Artificial Intelligence (AI)? Discuss brief history of AI with Chronological development. [2+6]
2. Why searching is necessary in AI? Explain about the role of production system with suitable example. [2+6]
3. What is horn clause? Differentiate between Depth First Search and Breadth First Search. [1+7]
4. Explain backward chaining with suitable example and compare with forward chaining. [4+4]
5. Why do we need FOPL? State any three rules of inference. How can we make the machine with learning capacity? [2+3+3]
6. Define Boltzmann Machine. How knowledge can be represented using semantic network? Explain with suitable example. [1+7]
7. What is Machine Learning? What is Fuzzy Logic? Explain the Fuzzy Inference with suitable example. [2+6]
8. Differentiate declarative knowledge and procedural knowledge. Explain the architecture of expert system. [2+6]
9. What is the role of perceptron in neural network? Explain about bakpropagation algorithm. [3+5]
10. What is Natural Language Processing (NLP)? Discuss the different issues related with NLP with example. [2+6]

Exam.	Regular		
Level	BE	Full Marks	80
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Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Define AI. When a machine is said to be passed Turing test? Give any two examples of constraint satisfaction problem. [2+5+1]
2. Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. WRONG + WRONG = RIGHT. Explain the steps that you have followed. [5+3]
3. Differentiate between informed and blind search. How depth search is different to breadth first search. Compare with evaluation parameters. [4+4]
4. All oversmart persons are stupid. Children of oversmart persons are naughty. Ram is children of Hari. Hari is oversmart. Show that Ram is naughty. Using FOPL based resolution method. [8]
5. Explain the step involved in conjunctive normal form (CNF) with suitable example. [8]
6. What is semantic net? Explain with suitable example. [8]
7. What is machine vision? Discuss about the algorithm of Genetic Algorithm. [2+6]
8. What is neural network? Explain back-propagation algorithm learning.. [4+4]
9. What is an Expert System? Explain the steps of an Expert System development. [4+4]
10. Define machine translation in NLP. Explain the challenges of machine translation. [1+7]

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2069 Poush

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
✓ Attempt All questions.
+4] ✓ The figures in the margin indicate Full Marks.
✓ Assume suitable data if necessary.

- 4] 1. What is artificial intelligence (AI)? Explain the Turing and its importance in artificial intelligence. [7]
- 2] 2. Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. [7]
2] RIGHT + RIGHT = WRONG
Show all the step of solving through constrain satisfaction problem.
3. What is blind search? Explain depth first search algorithm with an example and compare it with breadth first search. [9]
4. Explain all the steps involved in conversion of conjunctive normal form (CNF) with an example. [10]
5. What is a frame? Explain it with suitable example. [7]
6. What is genetic algorithm (GA)? Explain operator of GA with its importance. [10]
7. What is neural network? Differentiate between supervised and unsupervised learning. [6]
8. What is expert system? Explain general architecture of an expert system. [8]
9. What is natural language processing (NLP)? Explain different steps of NLP. [8]
10. Write short notes on: (any two) [4×2]
 - a) Horn clause
 - b) Machine vision
 - c) Perceptron

Exam. Level Programme Year / Part	Regular (2066 & Later Batch)		
	BE	Full Marks	80
	BCT	Pass Marks	32
	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Discuss any two fields of your daily life where artificial intelligence has been applied. (7)
2. Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. $LOGIC + LOGIC = PROLOG$
Show all the step of solving through constraint satisfaction problem. (7)
3. Discuss the hill-climbing search algorithm along with problems associated with it and discuss their solutions. (9)
4. Given premises: Every American who sells weapons to hostile nations is a criminal. The country Abc is enemy of America. All of the missiles in Abc were sold by John. John is an American.
Proof: John is a criminal. (10)
5. What are the different knowledge representation models? Discuss semantic nets with an example. (7)
6. What is Fuzzy learning? Explain with a practical example. (4)
7. Explain the learning framework with suitable example. (6)
8. What is a Hopfield Network? Explain all the steps involved in the Hopfield Network with suitable example. (8)
9. Explain different steps of expert system development with an example. (8)
10. What is a natural language processing? Explain it. (6)
11. Write short notes: (any two) (4 x 2 = 8)
 - i. Skolemization
 - ii. Machine vision
 - iii. Human Brain verses Neural Network
